

| Image    | Final answer   | Confidence | Two-stage result                            |                                         | Final probability |              |              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------|------------|---------------------------------------------|-----------------------------------------|-------------------|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                |            | Stage 1:<br>Batch 3 or not                  | Stage 2:<br>Batch 1 or 2                | Batch 1           | Batch 2      | Batch 3      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0eryguqq | <b>Batch 3</b> | High       | <b>Batch 3: 86.3%</b><br>Not Batch 3: 13.7% | —                                       | 6.4%              | 7.3%         | <b>86.3%</b> | Mainly fine surface detail in the InLens image.<br><ul style="list-style-type: none"> <li>InLens local contrast at 0.2 <math>\mu\text{m}</math>: 1.8 SD above the Batch 3 average, still within the normal Batch 3 range</li> <li>Texture fairly even at 0.05 <math>\mu\text{m}</math> and 0.2 <math>\mu\text{m}</math>: about 1.2 SD above average</li> <li>Contrast about as strong left-to-right as top-to-bottom: 2.4 SD below average</li> <li>BSE small dark notches at a typical Batch 3 rate: 1.4 SD above average</li> </ul> Every feature the model relies on falls within the Batch 3 range, and none points to Batch 1 or 2. |
| fhwrjtet | <b>Batch 3</b> | High       | <b>Batch 3: 86.3%</b><br>Not Batch 3: 13.7% | —                                       | 6.6%              | 7.1%         | <b>86.3%</b> | Fine InLens surface detail, weighed against how the silicon and graphite are arranged.<br><ul style="list-style-type: none"> <li>Points to Batch 3: InLens local contrast 2.3 SD above average at 0.2 <math>\mu\text{m}</math> and 2.7 SD at 0.05 <math>\mu\text{m}</math>; BSE dark-notch texture 1.3 SD above average</li> <li>Points away: silicon spread less evenly than usual (2.1 SD below average); graphite flakes strongly lined up in one direction (5.7 SD above average). Both are more like Batch 1</li> </ul> The texture evidence outweighs the two features that point elsewhere.                                       |
| 4hq27w4c | <b>Batch 1</b> | Low        | Batch 3: 10.9%<br><b>Not Batch 3: 89.1%</b> | <b>Batch 1: 50.7%</b><br>Batch 2: 49.3% | <b>45.2%</b>      | 43.9%        | 10.9%        | Stage 1, not Batch 3: the ETD fine texture has clearly fewer sharp features than Batch 3.<br><ul style="list-style-type: none"> <li>Sharp corners: 3.0 SD below the Batch 3 average</li> <li>Straight edges: 2.7 SD below</li> <li>Soft corners: 2.3 SD below</li> <li>Gentle curves: 2.2 SD below</li> <li>BSE agrees: sharp corners 2.1 SD below</li> </ul> Stage 2, Batch 1 or 2: 50.7% Batch 1 against 49.3% Batch 2; this split is not statistically supported.                                                                                                                                                                     |
| y59rxmxl | <b>Batch 1</b> | Low        | Batch 3: 10.9%<br><b>Not Batch 3: 89.1%</b> | <b>Batch 1: 51.6%</b><br>Batch 2: 48.4% | <b>46.0%</b>      | 43.1%        | 10.9%        | Stage 1, not Batch 3:<br><ul style="list-style-type: none"> <li>Strongest signal: InLens texture is much more even left-to-right than top-to-bottom, 5.2 SD above the Batch 3 average, the most extreme measurement in this set</li> <li>ETD and BSE fine texture has fewer gentle curves and sharp corners (2.0 SD and 2.1 SD below average)</li> <li>Points toward Batch 3: pore space changes more than usual from top to bottom (2.7 SD above average), but the texture evidence outweighs it</li> </ul> Stage 2, Batch 1 or 2: 51.6% Batch 1 against 48.4% Batch 2; this split is not statistically supported.                      |
| fspqbkxl | <b>Batch 2</b> | Low        | Batch 3: 10.9%<br><b>Not Batch 3: 89.1%</b> | Batch 1: 47.5%<br><b>Batch 2: 52.5%</b> | 42.3%             | <b>46.8%</b> | 10.9%        | Stage 1, not Batch 3: the fine texture is smoother and flatter than in Batch 3, on both detectors.<br><ul style="list-style-type: none"> <li>ETD gentle curves: 2.2 SD below the Batch 3 average</li> <li>BSE sharp corners: 1.8 SD below; BSE gentle curves: 1.6 SD below</li> <li>Flat areas and small dark specks more common: 1.8 SD above average in ETD, 1.5 SD above in BSE</li> </ul> Stage 2, Batch 1 or 2: 47.5% Batch 1 against 52.5% Batch 2; this split is not statistically supported.                                                                                                                                     |
| soo2ax3r | <b>Batch 2</b> | Low        | Batch 3: 18.0%<br><b>Not Batch 3: 82.0%</b> | Batch 1: 49.1%<br><b>Batch 2: 50.9%</b> | 40.2%             | <b>41.7%</b> | 18.0%        | Stage 1, not Batch 3: the least certain of the exclusions; the evidence is mostly about the material itself.<br><ul style="list-style-type: none"> <li>Mix of silicon particle types differs from Batch 3: one type much rarer (2.7 SD below average), another much more common (2.7 SD above)</li> <li>Silicon looks darker compared with graphite (2.2 SD below average)</li> <li>InLens surface less smooth at both scales (1.5 SD below average)</li> </ul> Stage 2, Batch 1 or 2: 49.1% Batch 1 against 50.9% Batch 2; this split is not statistically supported.                                                                   |

## Limitations

- Batch 1 vs Batch 2:** with this data, the images don't give a reliable way to separate them, so we report both as possible instead of guessing confidently.
- Imaging settings:** some fine-texture features may partly reflect how the image was taken (detector noise, focus), not only the material. The features about the material itself are more physically meaningful: how evenly the silicon is spread, the mix of silicon particle types, and how the graphite is aligned.